IntelliHire LangGraph graphs

Every LangGraph graph in the IntelliHire backend, as Mermaid flowcharts. Each diagram was generated from the compiled graph itself with get_graph().draw_mermaid(), so it shows the real nodes and edges in the code.

Solid arrows are fixed edges. Dotted arrows are conditional edges: the routing function after a node picks one of them at run time. A node with an arrow back to itself can retry (for example, interpret trying again when it found no usable time window, or human_review waiting again when an approval is sent for a draft that has not passed the guardrails).

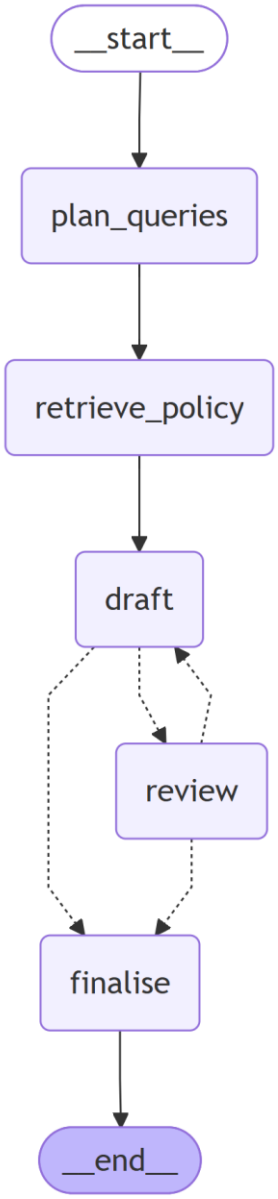
Contents

1. Job description agent 2
2. CV screening 3
3. Candidate scoring 4
4. Interview scheduling 5
5. Interviewer co-pilot 6
6. Q&A analysis and requirement mapping 7
7. Consolidated assessment 8
8. CV claim checks 9
9. Interview score 10
10. Interview quality 11
11. Candidate feedback composer 12
12. Skeleton graph (reference example) 13

1. Job description agent

Source: `backend/app/agents/jd/graph.py`

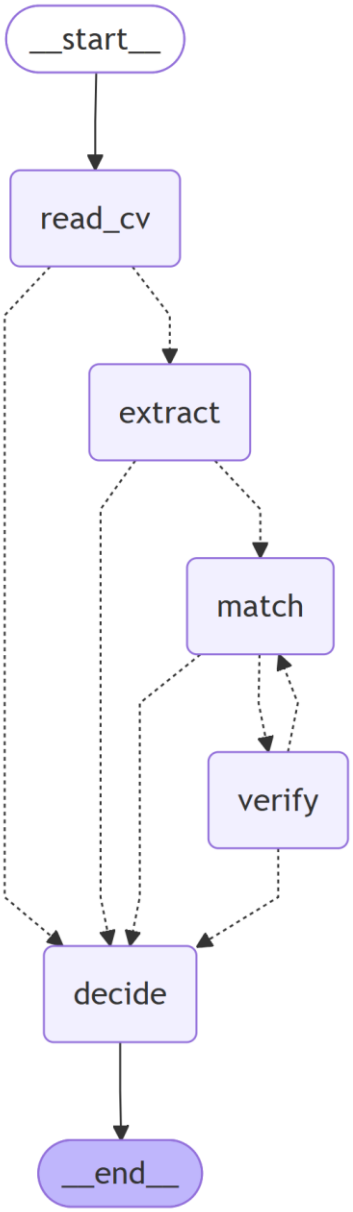
Drafts or revises a job description from the job profile and the company policies. Plans policy queries, retrieves passages, drafts with structured output, then a code review step can send the draft back for a redraft. Also serves the chat mode.



2. CV screening

Source: `backend/app/agents/screening/graph.py`

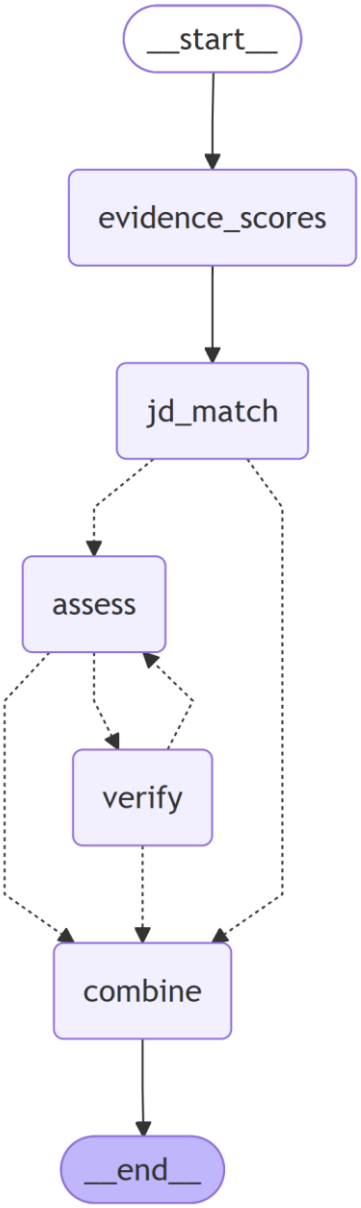
Reads one CV (with redaction), extracts the candidate's evidence, matches it to the job requirements, verifies every quote against the stored CV text and decides: ranked, ruled out on a must-have, or set aside.



3. Candidate scoring

Source: `backend/app/agents/scoring/graph.py`

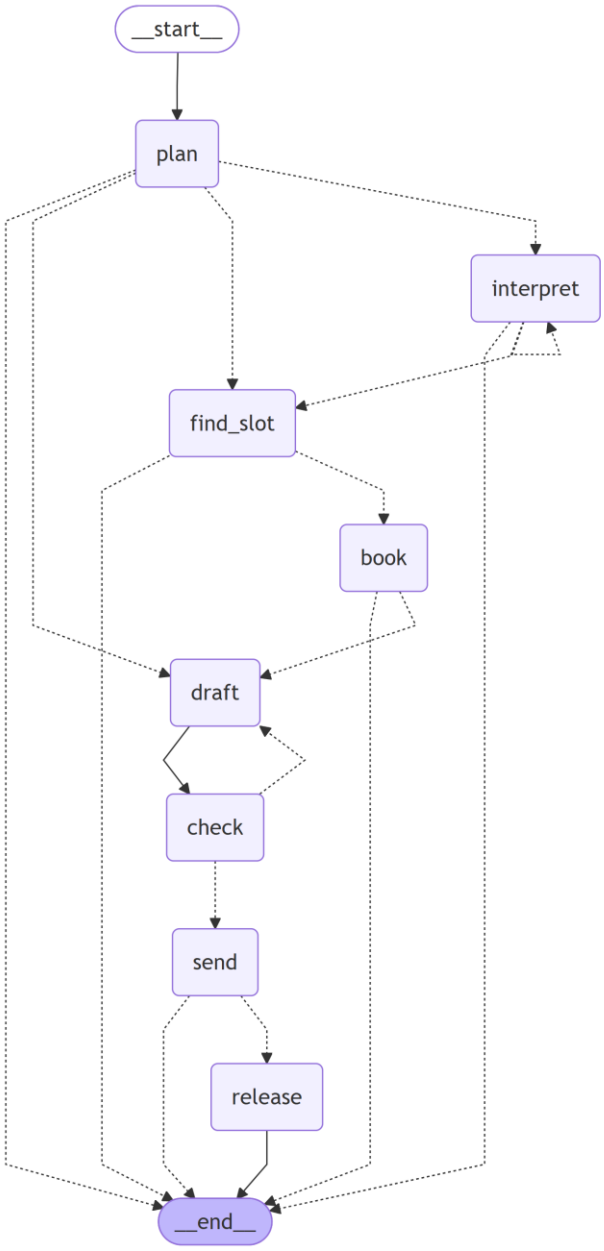
Starts each dimension from the verified screening evidence, adds an ATS-style description match, lets the AI propose a capped, quoted adjustment, verifies it, and combines the weighted score.



4. Interview scheduling

Source: `backend/app/agents/scheduling/graph.py`

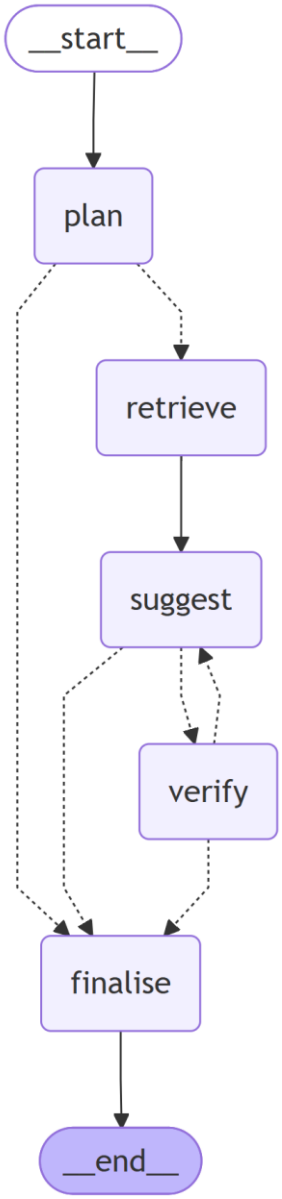
Schedules, reschedules or cancels an interview. Interprets the request, finds a free slot, books the calendar event, drafts and checks the email, and sends it through the MCP server. If sending fails, release undoes the booking.



5. Interviewer co-pilot

Source: `backend/app/agents/copilot/graph.py`

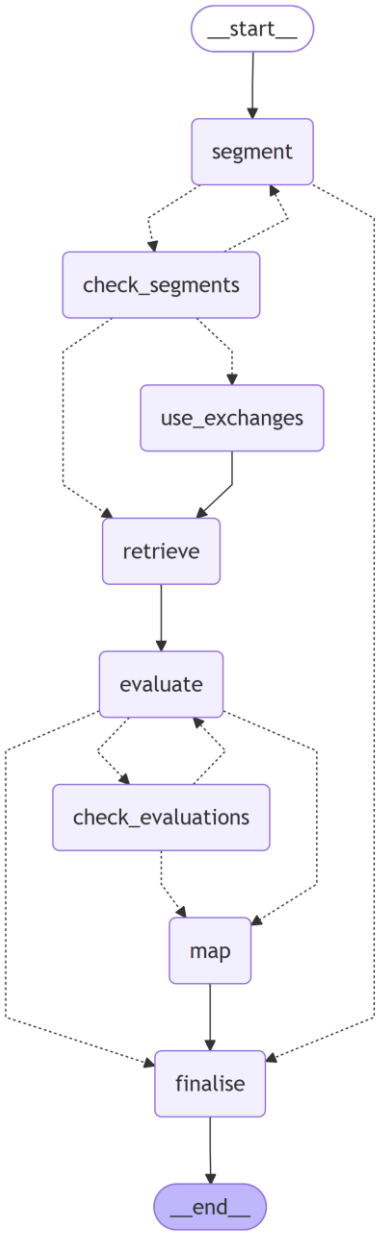
During a live interview, plans which requirements still need probing, retrieves from the reference library, suggests questions, and verifies quotes and citations before the suggestions are saved.



6. Q&A analysis and requirement mapping

Source: `backend/app/agents/analysis/graph.py`

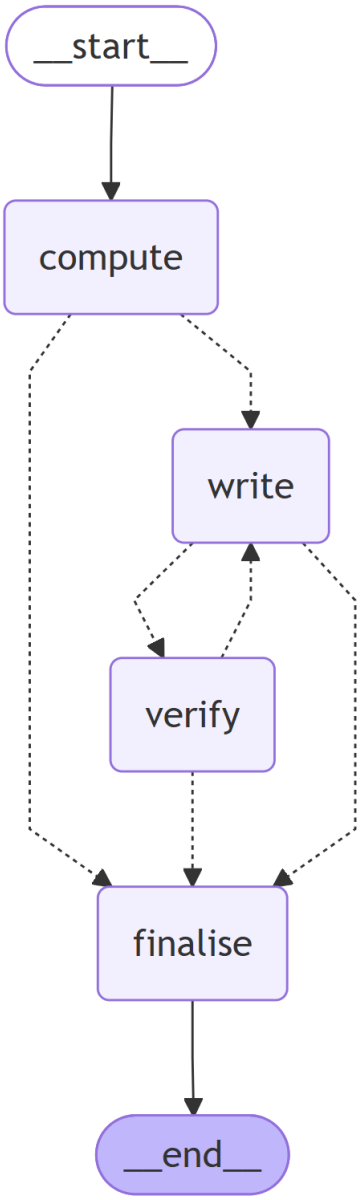
After an interview, splits the transcript into question-and-answer exchanges, checks the split, retrieves references, evaluates each answer, checks the evaluations and maps them to the job requirements.



7. Consolidated assessment

Source: `backend/app/agents/assessment/graph.py`

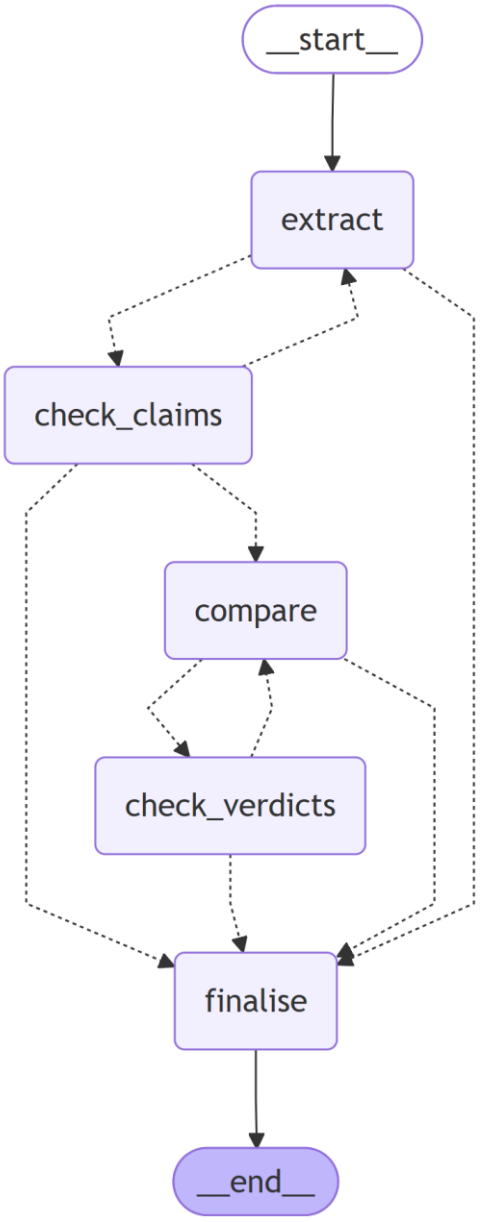
Computes the code-weighted position with the must-have cap, has the AI write the explanation of the findings, and verifies them. The AI never states a hiring verdict.



8. CV claim checks

Source: `backend/app/agents/claims/graph.py`

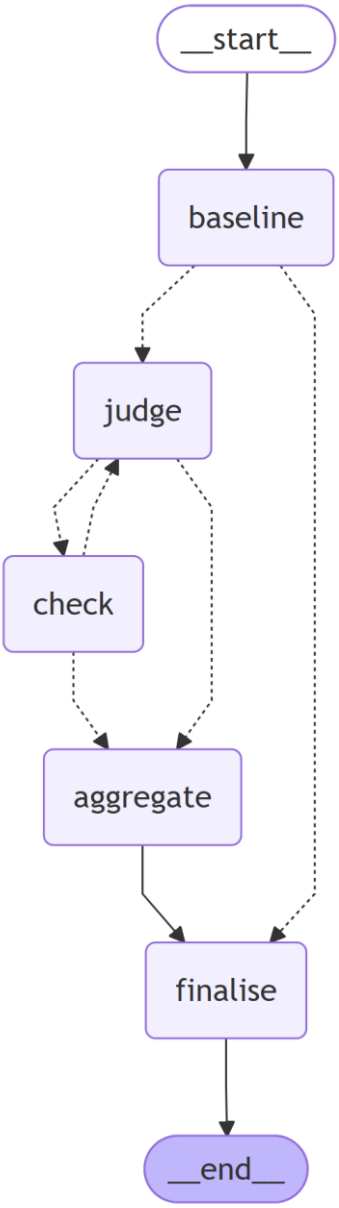
Extracts the claims a CV makes, checks them, compares them with what the candidate said in the interview, and checks the verdicts before they are saved.



9. Interview score

Source: `backend/app/agents/interview_score/graph.py`

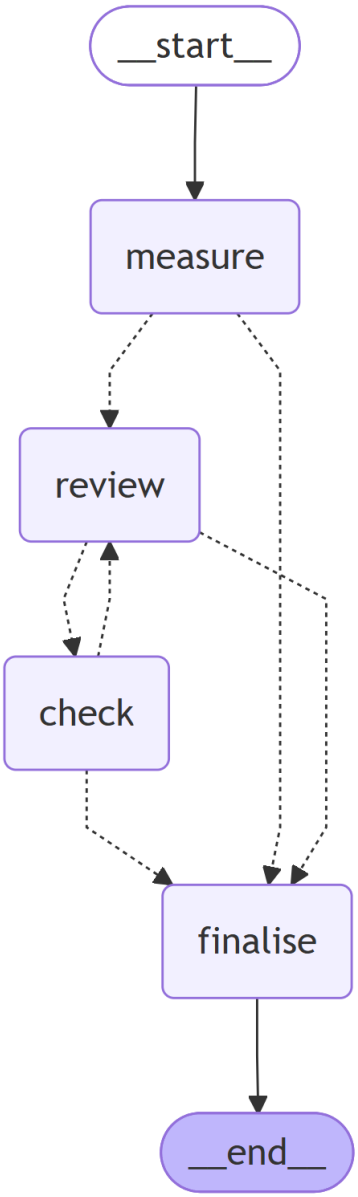
Builds a baseline from the analysis, lets the AI judge the answers, checks the judgements and aggregates them into the interview score.



10. Interview quality

Source: `backend/app/agents/interview_quality/graph.py`

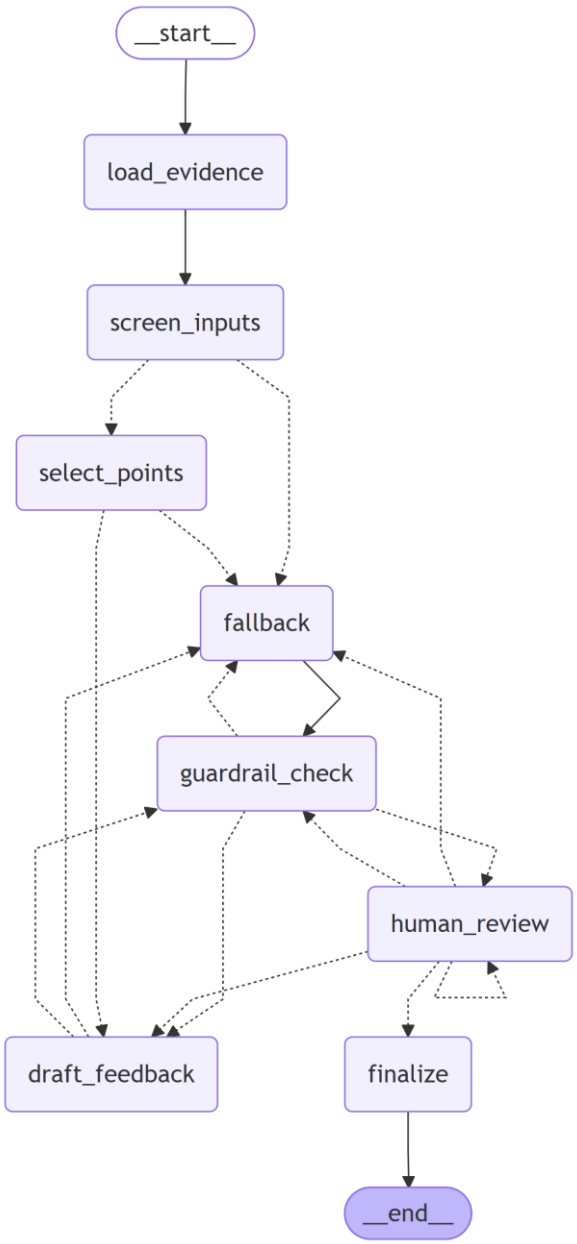
Computes metrics on the interview from the transcript turns, the Q&A pairs and the requirements (compared with other interviews for the same job), lists the issues, has the AI review them, and checks the review text.



11. Candidate feedback composer

Source: `backend/app/agents/feedback/graph.py`

Writes feedback for a candidate who was not selected. Loads stored evidence, screens the inputs, picks the points in code, drafts, and runs the rule and AI-judge guardrails. human_review is an interrupt: the graph waits for a person to approve, edit or redraft before finalize. Any failure goes to the fixed-wording fallback.



12. Skeleton graph (reference example)

Source: `backend/app/agents/skeleton.py`

The minimal worked example every feature graph copies: validate, transform, finalise.

